FOUNDATION
OPENHW™
PROVEN PROCESSOR IP

Driving European Digital Autonomy and Leadership through RISC-V

How can Europe capture the €790 billion semiconductor market?

Flo Wohlrab
Head of
OpenHW Foundation

Charley Mann
Comms Lead TRISTAN
Content Lead
OpenHW Foundation

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OpenHW Foundation

Non-profit hosted in Belgium by the Eclipse Foundation (AISBL)

Open source, permissively licensed, production-ready cores already used in EU projects

World leader in industrial-grade RISC-V core development

Global partners from industry and academia

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OpenHW Foundation Membership

cea list

THALES

10x ENGINEERS

AINEKKO.

Barcelona Supercomputing Center
Centro Nacional de Supercomputación

CAPABILITIES LIMITED

CIRCUIT SUTRA TECHNOLOGIES

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POLYTECHNIQUE MONTRÉAL
WORLD-CLASS ENGINEERING

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OPENHW FOUNDATION
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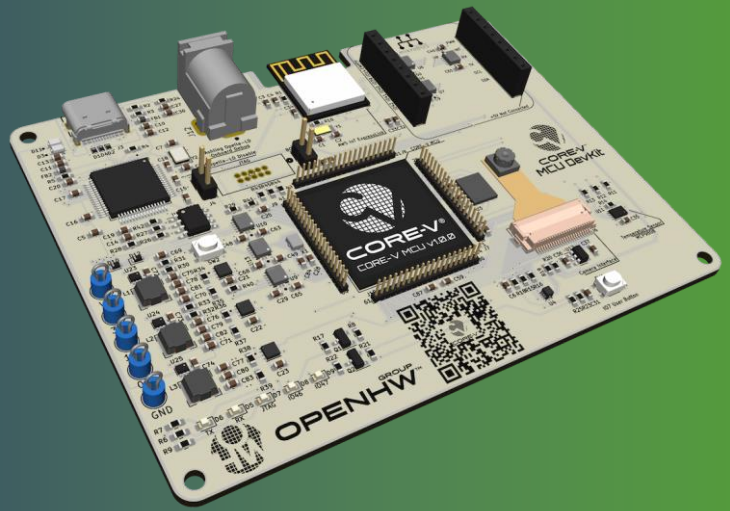
CORE-V Family: A Rich IP Portfolio

CORE-V is a family of RISC-V based open source cores with processor subsystem IP, tools, and software designed to accelerate innovation in electronic systems. These cores are optimized for both silicon and FPGA implementations, enabling rapid design and manufacturability of SoCs.

- **CVA6**: High-performance 64/32-bit cores for Linux-based systems and safety-critical applications.
- **CVW / CV-Wally**: A textbook-based 5-stages 64/32-bit core configurable to support all the standard RISC-V extensions, .virtual memory, PMP, and the various privileged modes and CSRs.
- **CVE4**: 32-bit embedded-class cores optimized for IoT, edge computing, and consumer electronics.
- **CVE2**: Ultra-efficient processors for deeply embedded control applications.
- **CVA6 Platform**: Vendor-neutral software validation platform for various FPGA configurations, including AWS cloud solutions.



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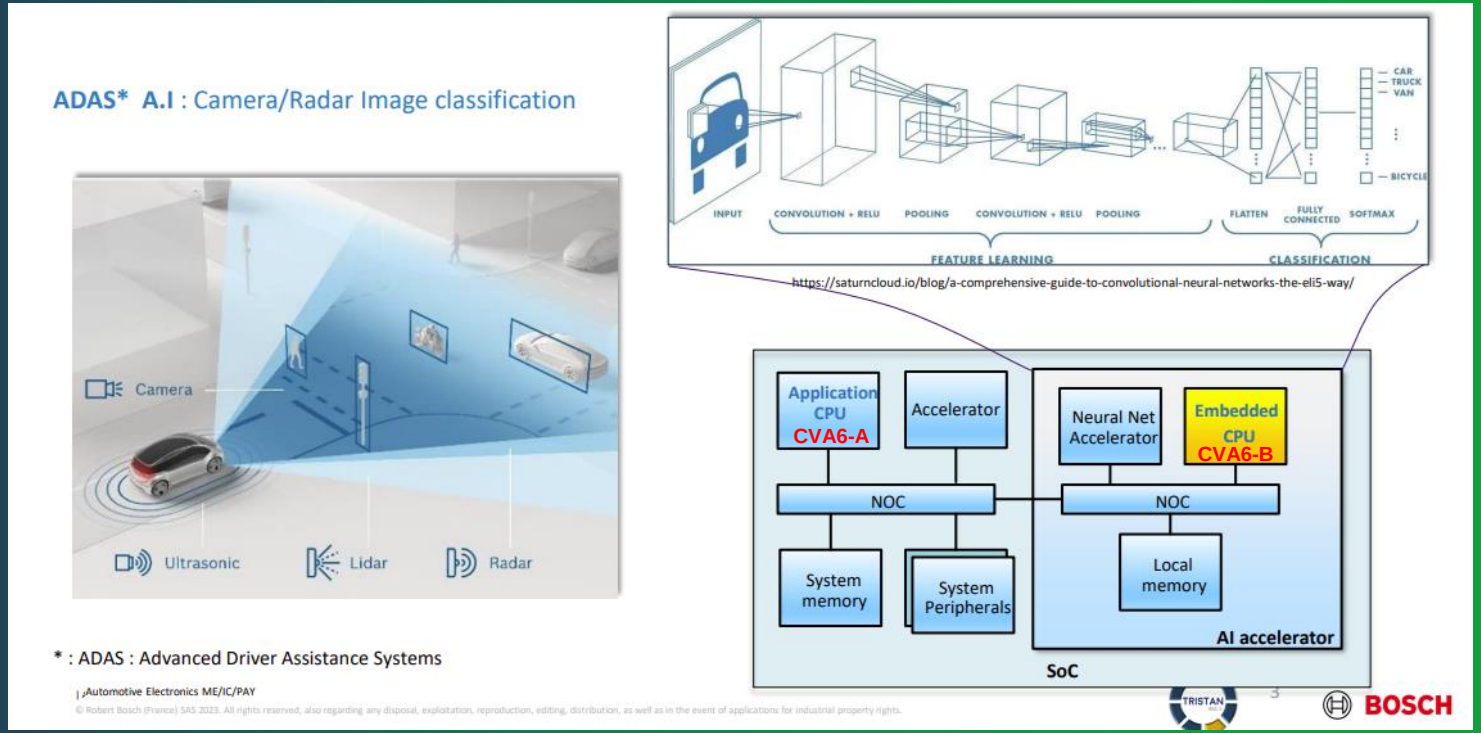
RISC-V in Automotive: Bosch Innovations

ADAS for object detection powered by OpenHW CVA6 RISC-V

2 Cores, different config.

Bosch via TRISTAN



Source: https://www.youtube.com/watch?v=Hfj7wsad1tA

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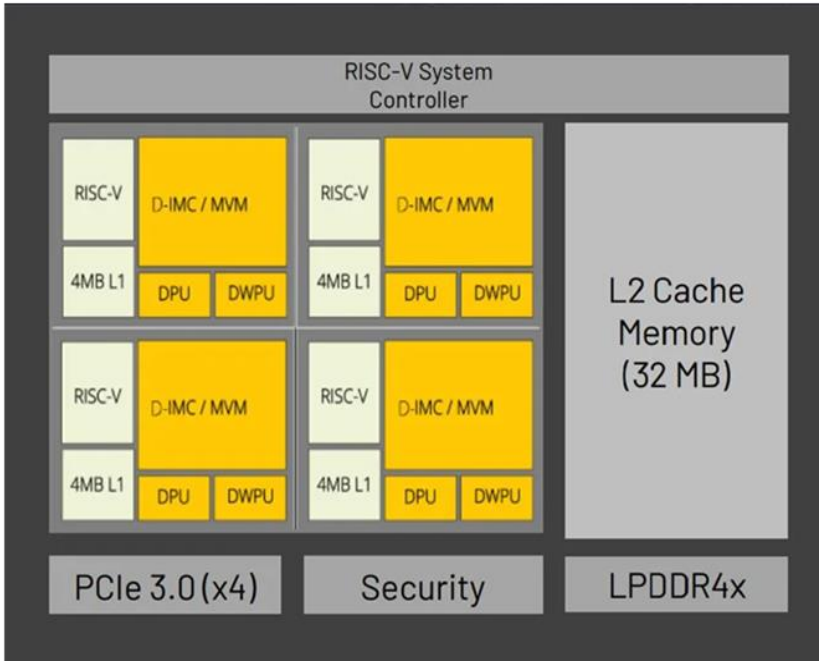
AI at the Edge - Made in Europe

AXELERA
ARTIFICIAL INTELLIGENCE

Compute in Memory -
driven by Open Source CVA6 RISC-V Cores

Quick adjustment, commercial viability

You can buy the kits now with Open Source RISC-V



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Semiconductors: A Global Industry

The Oil of the Digital Age

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Why do we need semiconductors?

Automation

Computing

Communications

Energy Management

Healthcare

Sensing

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Semiconductor global market

$790 Billion

2025

$1 Trillion

2026

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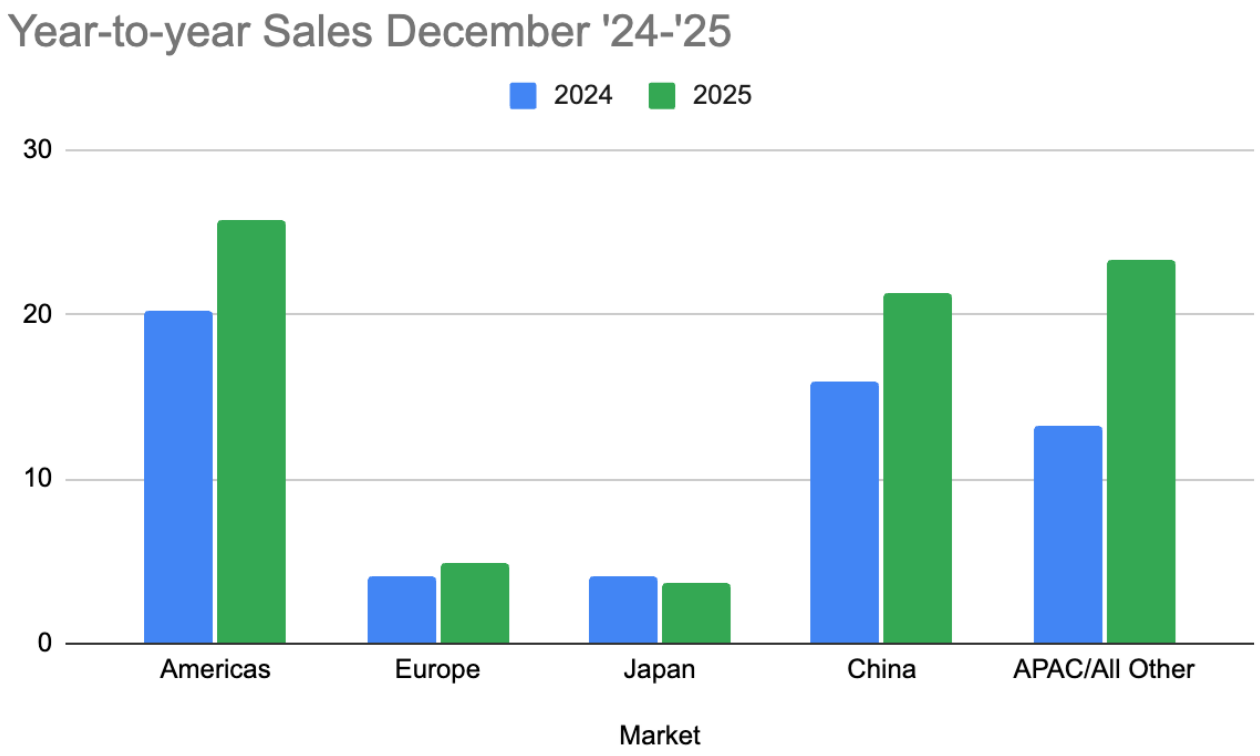
Semiconductor global market



SIA/WSTS Global Semiconductor Sales Table

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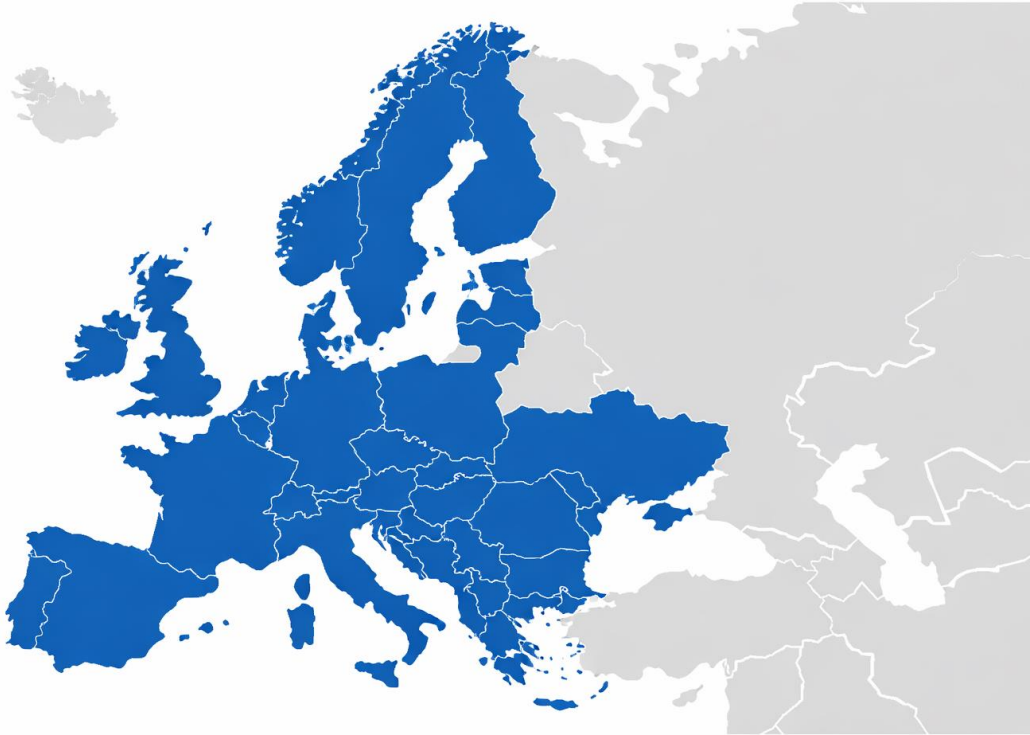
Europe on the periphery

The EU accounted for just

6%

of the global semiconductor market in 2025



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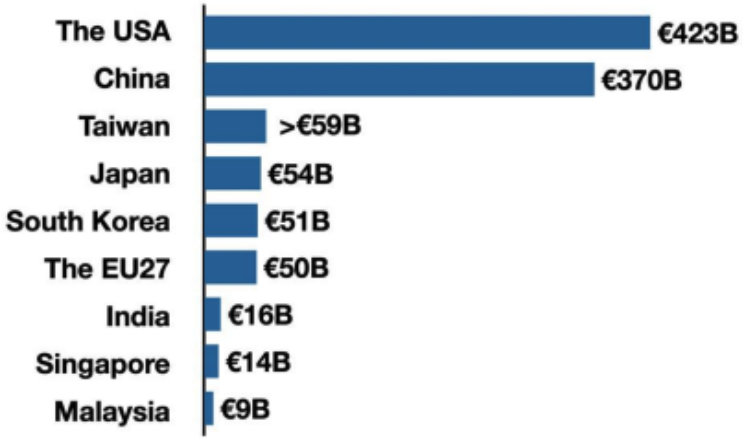
Investment in Semiconductors

The diagram below provides an overview of confirmed ongoing investments along the semiconductor value chain[7], by country or region, as of May 2025. Only individual projects exceeding € 1 billion are included.

Figure 15: Confirmed Semiconductor Investments Over € 1 Billion by Country/Region – May 2025



Source: DECISION Etudes & Conseil

https://www.decision.eu/wp-content/uploads/2025/08/ICOS-D2.3-Deliverable_final-PUBLIC.pdf

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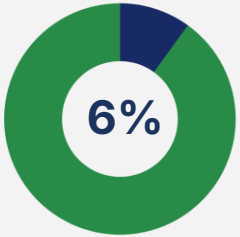
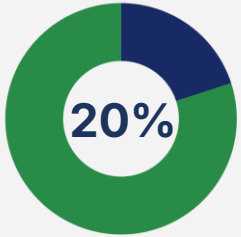
Why do we need RISC-V?

$790B

Semiconductor Market Value 2025



EU Market Share Today



EU Market Share by 2030

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A Vision for the Future

Toward an Autonomous, Trustable & Scalable RISC-V Stack for Europe

Functional Safety & Trustable Software

Eclipse TSF

Application Building Blocks

SDV Eclipse Software Defined Vehicle

eclipse iot

Developer Tools

eclipse

THEIA

Open VSX

ECLIPSE Cloud DevTools

Embedded RTOS

oniro

ECLIPSE THREADX

RISC-V Cores & Toolchains

OPENHW FOUNDATION PROVEN PROCESSOR IP

AI Enablement

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A Vision for the Future

Digital autonomy

≠

Isolationism

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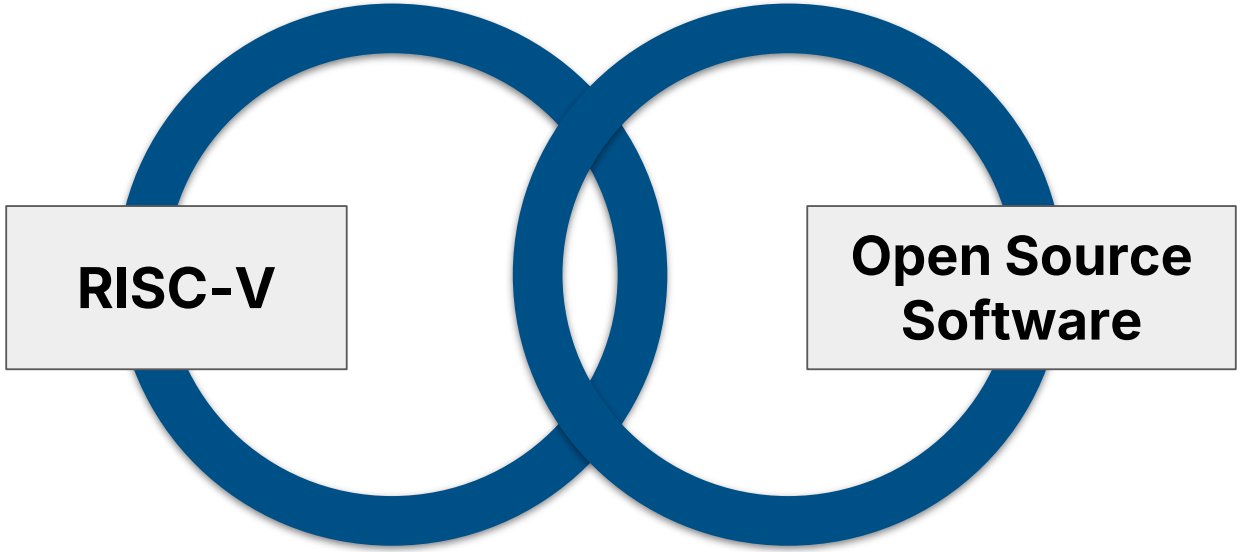
Stronger Together



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RISC-V in Europe

Building the Infrastructure for European Success

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RISC-V in Europe

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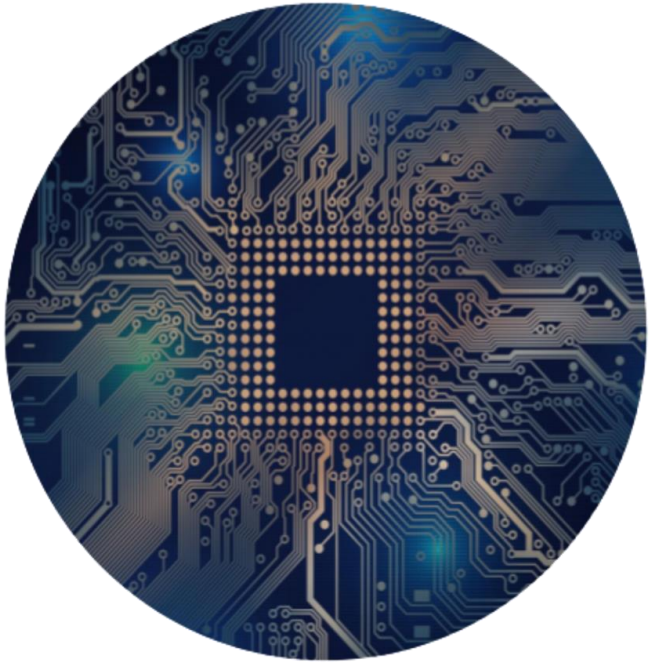
TRISTAN



The TRISTAN Project

€50M+ initiative to mature and industrialize the European RISC-V ecosystem.

46 partners including NXP, Thales, and ETH Zurich.

Funded by the European Union

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RISC-V in Europe

The Silo Problem

Europe has world-class IP, but it's hidden in silos:

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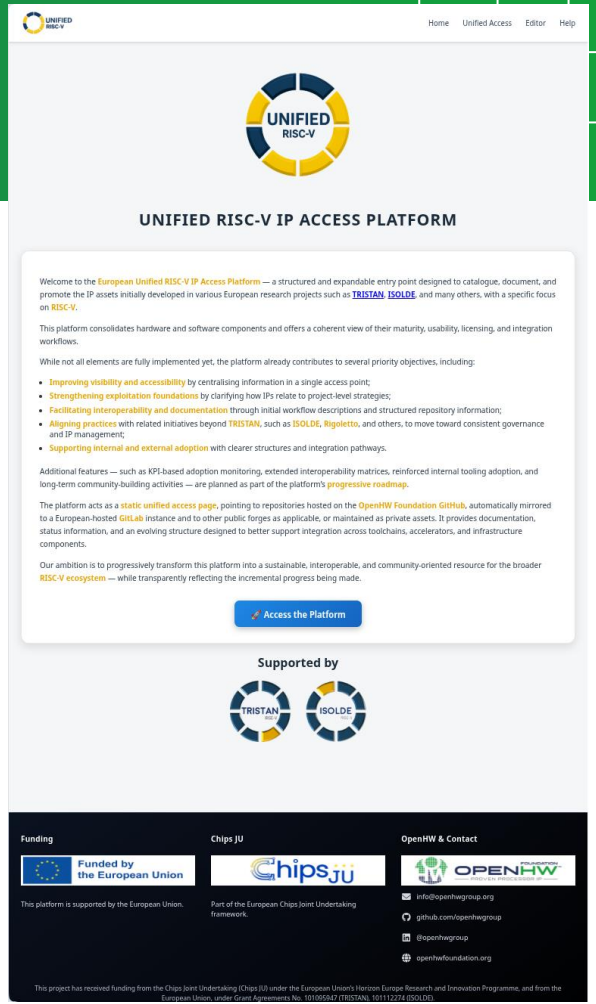
Unified RISC-V IP Access Platform

- **Available at** https://openhwgroup.github.io/tristan-isolde-unified-access-page
- **Embedded at** TRISTAN web page at https://tristan-project.eu/uap

Overview

- **Welcome page** – welcome page encompassing the role of the UAP as the reference for developed RISC-V IPs developed under EU projects (TRISTAN, ISOLDE and others).
- **Platform page** – comprising a dynamic table, with the latest IPs information as available under the Virtual Repository.
- **Editor page** – to edit or update IP and project entries.
- **Help page** – guide on how to contribute to the UAP or its Virtual Repository catalogue of IPs, *currently under construction*.



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Unified RISC-V IP Access Platform

Name	Category	License	Status	Description	Project
Compression and decompression of digital waveforms	Core	TBD	In progress	Custom instructions for CV32E40X core to improve the performance of real-time compression and decompression of digital waveforms	TRISTAN
CVA6	Core	Solderpad	In progress	A configurable family of RISC-V application/embedded cores targetting FPGA and ASIC technologies	TRISTAN
CVE2	Core	Solderpad	In progress	A single-issue 2-stage pipeline embedded class of RISC-V CPUs	TRISTAN
Extensions to the micro-architecture of CV32E40P core	Core	Solderpad	In progress		TRISTAN
Hypervisor	Core	Solderpad	Released	Hypervisor support for CVA6 complying with the RISC-V hypervisor extension specifications	TRISTAN
Riviera: RISC-V ISA Extensions for NFC Applications	Core	LA_OPT_NXP Software License	Design and Verification completed	CV-X-IF compliant RISC-V Co-processor for a NFC Receiver decoder custom DSP acceleration	TRISTAN
RVB / RVP Standard Extensions support for CV32E40P core	Core	Solderpad	In progress	Development of light-weight RISC-V Instruction Set Architecture (ISA) extensions to improve the energy efficiency of low-bit-width mixed-precision integer arithmetic	TRISTAN
RVV coprocessor for CVA6	Core	Solderpad	Released	RVV (vector) co-processor for CVA6 with support for low precision integer arithmetic (down to 8-bit vector data types) and multi-precision floating-point operations	TRISTAN
SCAIE-V custom instruction interface for CVA6	Core	Apache 2.0	In progress	SCAIE-V is a portable and scalable hardware interface for easily adding custom instructions to processors ranging from microcontrollers to application-level cores	TRISTAN

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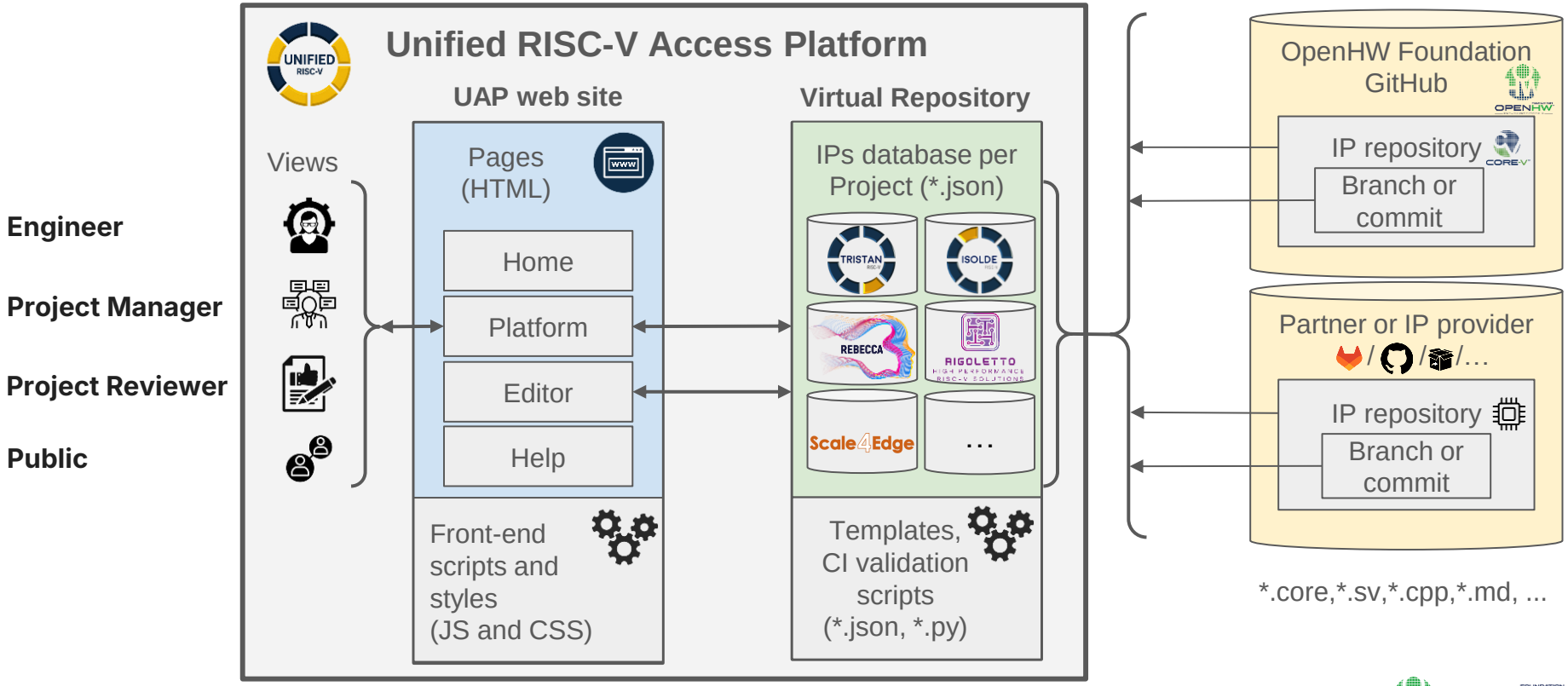
Unified RISC-V Access Platform infrastructure



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The Virtual Repository features

Decentralized

IPs remain on partner servers. We index the metadata, not the IP itself.

Ex.: For OpenHW IPs, it points to their GitHub repos

Discovery

One single portal to find verified RISC-V cores, accelerators, IP generators and software across multiple projects.

Sovereignty

Hosted following FAIR[1] data principles access and backed-up on European infrastructure.

[1] - https://www.go-fair.org/fair-principles/

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Join the RISC-V Revolution: Adopt IP and Contribute Back

Thank you

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